Cybersecurity Threats in Cloud Computing: Analysis of Threat Modeling and Risk Assessments

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Abstract

[fill in what this review paper is going to be about]

Introduction

Cloud computing is the on demand delivery of computing resources. Specifically servers, data storage, networking, application development tools, and software;over the internet on a pay-as-you-go basis. It allows individuals and organizations to utilize remote servers housed in large data centers without owning or managing physical hardware. Cloud computing's flexibility and scalability have made it integral to a wide variety of applications, from everyday tasks like streaming media and email access to business processes requiring vast amounts of computing power and storage. By leveraging cloud services, users can scale resources quickly, enable collaboration, and leverage emerging technologies without having to make significant upfront capital investments. As the adoption of cloud computing increases among organizations, security has become a major concern with respect to cybersecurity threats. But as cloud computing becomes more widely used, cybersecurity is becoming an important matter. As more businesses shift their activities to the cloud, they must contend with a constantly changing array of security risks. These risks include data outsourcing that leads to malicious attacks, human error leading to data loss, account hijacking, DDoS attacks, misconfiguration, API vulnerabilities, and malware/ransomware. All of these pose serious risks to sensitive information and businesses. (Thesis) Although there are risk models in place, a lot of these traditional risk models struggle to face the evolving threats. Encrypted data limits functionality, Log audits are insufficient and can be tampered with,auditors themselves can be compromised. Among these difficulties is the urgent need for strong cybersecurity measures that can protect cloud systems from evolving threats. These limitations require effective risk model assessments such as the fuzzy set analysis. This paper discusses the various security controls in

place, how threat modeling helps identify these threats, and how risk assessments help identify which model is the best and which threats need most attention. Our review concludes Stride and fuzzy logic help improve risk prioritization the most.

Cybersecurity Threats in Cloud Computing

Research by Awodele et al. (2024) evaluated cloud security threats and assigned risk values based on likelihood and impact. Their analysis found that data breaches and insider threats ranked among the most severe threats in cloud computing.Some of the common threats in cloud computing are data breaches which are one of the most serious threats to cloud systems, caused by the loss of encryption keys and the disclosure of private information. They are positioned as high-risk to negatively impact organizations. Account Hijacking is also a significant risk, especially if passwords are overused or weak. Given that impersonation and unauthorized access to login credentials are common attack routes. There's also Insider threats. In many situations, malicious insiders pose a significant risk, particularly when cloud services are being threatened, especially when proper inspection is lacking. Because insiders have privileged access within an organization, it can be challenging to identify insider threats. Lastly, Insecure Interfaces and APIs. Because attackers can take advantage of poorly designed or exposed interfaces and APIs, this poses a significant threat to cloud cybersecurity. Under factors like misplaced encryption keys, illegal access to login information, and misuse of cloud services.

- **Security Controls for Mitigation:** Encryption, Firewalls, Access Controls, MFA, IDS, etc.

- **Limitations of These Controls:** Why traditional security controls alone aren't enough (e.g., encrypted data reduces system functionality, log audits can be tampered with).

Threat Modeling & Risk Assessment

Security control alone is not effective enough to battle against threats. Establishing a blueprint before executing a practice can help improve the chances of mitigating threats immensely. Which is why threat models exist. Threat modeling as described by Bodeau, McCollum, and Fox (2018) is a structured process used to uncover, examine, and mitigate cybersecurity threats at various levels, including system, business, organizational, and enterprise levels. At the system levels, it is used to guide security control decisions in operations and design. At the organizational level, it reflects the organization's assumptions about its threat environment and influences risk management decisions, including SLAs with service providers like cloud services. At more aggregated levels, threat modeling facilitates the sharing of threat intelligence and supports collaborative efforts like cyber wargames. Overall, it constitutes a valuable component of proactive risk analysis and serves to ensure security is set in throughout the system and organization life cycle.

Some of the popular techniques are STRIDE, for example, according to Naik et al. (2024) stride divides threats into six categories: Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege. STRIDE is effective in spoofing and tampering concerns. It mainly focuses on the part where data is moving in the cloud which will help discover any vulnerabilities much easier (pg.271). Attack trees are used to determine how an attacker would attempt to compromise a system. They make it simpler to evaluate risk or launch simulated cyberattacks (also known as cyber wargaming) by illustrating the various ways an attack could occur. PASTA (Process for Attack Simulation and Threat Analysis) is another technique that evaluates cloud security. Pasta has phases to fully cover a system.First phase is analyzing businesses applications then the hardware and software they are using. Second phase is a deeper dive into the data flow and data logs in these applications and detect vulnerabilities and possible threats. Third phase they would do an attack simulation to identify all the areas needing security control and then establish a security strategy for the business (pg.80). LINDDUN as well is a framework for privacy risk analysis.Gained recognition because of its

capacity to identify privacy issues in cloud systems through identifying privacy threats such as Linkability, Identifiability, Nonrepudiation, Detectability, Disclosure of information, Unawareness, and Noncompliance (pg.283).

To determine which threat model is the best, Awodele (2024) used a fuzzy set method as a risk assessment. Fuzzy set theory is a mathematical framework that can handle extreme ambiguity and vagueness in data.Their rationale for doing so is based on threat modeling, which makes it possible to examine a system's architecture, determine potential threats, and develop mitigation plans. The threat modeling process was conducted by sampling cloud infrastructures with the overall goal of helping experts develop a cyberthreat mitigation model that can be used in cloud environments. In their research applied fuzzy set theory to the analysis of cloud security risks' uncertainties and discussed various mitigation strategies,The fuzzy values range table (Table 1) was created to detect different risk tiers easily, from low to very high. The authors established specific value sets (0.1-1) and (1-5) . This lets them convey the challenges present in assessing cybersecurity hazards. The use of fuzzy values allows a clearer quantification of risks. This also aids organizations to better assess as well as react to these risks.

There were several threat modeling methods analyzed, including STRIDE, PASTA, LINDDUN, CVSS, Attack Tree, PnG, Security Cards, HTMM, QTMM, Trike, VAST Modeling, and OCTAVE. To support their analysis, the authors performed a comparative analysis of such methods of mitigation and assessed them according to 14 key attributes ranging from the quality of documentation, technical ability in detecting threats, efficiency in time, to usefulness in diverse cloud scenarios. Showing that STRIDE methodology as the optimal strategy.

		Threat Mitigation Techniques											
S/N	**Features**	**STRIDE**	**PASTA**	**LINDDUS**	**CVSS**	**Attack Tree**	**PnG**	**Security Cards**	**HTMM**	**QTMM**	**Trike**	**VAST Modelling**	**OCTAVE**
1	Documentation	5	4	4	2	2	2	1	4	1	2	4	2
2	Technical threat identification	5	5	3	5	5	5	5	2	5	2	2	3
3	Non-technical threat identification	5	3	3	5	5	3	3	4	5	4	4	4
4	General threat identification	5	5	4	5	5	3	3	5	5	4	3	3
5	Time consumption	2	1	2	3	3	4	3	4	3	1	1	4
6	Usage	5	1	3	4	4	5	3	3	4	3	5	4
7	Model maturity	5	3	4	3	3	3	2	4	5	3	3	4
8	Training/usage requirements	5	2	3	4	4	3	4	4	4	4	3	3
9	Business impact	1	5	2	3	1	3	1	2	3	3	4	4
10	Security properties	5	4	4	3	4	3	3	5	5	4	3	4
11	Threat classification	5	3	3	4	3	3	4	3	4	3	3	1
12	Stakeholders' input/collaboration	4	5	3	1	1	3	5	1	4	3	3	2
13	Threat prioritization	3	5	4	5	3	4	4	3	5	3	3	2
14	Reliability	4	4	3	4	3	3	1	2	4	4	4	4
15	**Total**	**59**	**50**	**45**	**51**	**46**	**47**	**42**	**46**	**57**	**43**	**42**	**44**

Not only is this method able to analyze risk assessments but also is able to analyze which cybersecurity threat poses the most risk. The table outlines the most common cybersecurity risks to cloud environments and simplifies them to data breaches, insider threats, account hijacking, DDoS attacks, misconfiguration, API vulnerabilities, and malware/ransomware as shown in table Table 3 is a quantitative assessment of risk for various scenarios and then generates an overall risk score for each danger. For instance, the threat of data loss from natural disasters has been given a fuzzy value of 0.75, which indicates a relatively high probability, while the value of 0.25, which indicates a lesser risk in this context, has been allocated to account hijacking by impersonation. In order to rank security measures, there is a total at the end of the table of the risk severity across all categories. This table is a key tool for converting qualitative risk data into taking action and ensuring that mitigation strategies match what needs to be prioritized.

Table 3. Analysis of cybersecurity threats using fuzzy set analysis

		Cybersecurity Threat Features										
S/N	**Causes**	**Data Loss**	**Data Breaches**	**Account Hijacking**	**Insecure Interfaces and APIs**	**Malicious Insiders**	**Insufficient Due Diligence**	**Abusive Use of Cloud Services**	**Shared Technology Issues**	**Identity Theft**	**Changes in Business Model**	**Lock -in**
1	Natural disaster	0.75	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.75	0.25
2	Loss of encryption key	0.75	0.75	0.50	0.75	0.25	0.25	0.25	0.25	0.25	0.25	0.25
3	Illegal access to login details	0.25	0.75	0.75	0.25	0.25	0.25	0.25	0.25	0.75	0.25	0.25
4	Undeserved operations conducted by users	0.75	0.75	0.75	0.25	0.75	0.25	0.25	0.25	0.25	0.25	0.25
5	Unknown cloud storage	0.25	0.75	0.25	0.25	0.75	0.75	0.25	0.25	0.25	0.25	0.75
6	Exposure to sensitive information	0.25	0.75	0.25	0.25	0.75	0.25	0.25	0.25	0.25	0.25	0.25
7	Unlawful acts of users	0.75	0.75	0.25	0.25	0.75	0.25	0.75	0.25	0.25	0.25	0.25
8	Usage of the same VM by multiple users	0.25	0.25	0.75	0.25	0.25	0.25	0.75	0.75	0.75	0.25	0.25
9	Impersonation	0.25	0.25	0.75	0.25	0.25	0.25	0.25	0.75	0.75	0.25	0.25
10	Operating system failure	0.75	0.75	0.75	0.25	0.25	0.25	0.25	0.25	0.25	0.75	0.25
11	Following internal security measures	0.75	0.75	0.75	0.25	0.25	0.75	0.25	0.25	0.25	0.25	0.25
12	**Total**	**11.5**	**13.5**	**12**	**6.5**	**9.5**	**4.5**	**7.5**	**4**	**4.5**	**4**	**3.5**

4. Why threat Models Fall Short (2 Pages)

- **Problems with Existing Models** (Probabilistic methods, Attack Trees, CVSS, etc.):

- Evidence from first link: Stride: STRIDE's limits in highly dynamic cloud infrastructures, where traditional threat models would not be as flexible. PASTA's effectiveness, the article noted that smaller businesses with fewer resources may struggle to implement it because of the substantial time commitment involved with its detailed nature.LINDDUN may not be as efficient in complex cloud systems with considerable data processing across various domains

  - Lack of flexibility for dynamic cloud environments.
  - High computational cost in large cloud infrastructures.
  - Poor handling of encrypted data and side-channel attacks.

- **Specific Case Study Example**: Show how a traditional model failed or struggled in a real-world scenario.

5. Improving Cloud Security with STRIDE (2 Pages)

- **Why STRIDE is the Best Threat Model:** Covers spoofing, tampering, repudiation, information disclosure, denial of service, and escalation of privilege.
- **Case Study Application:** Take a real life scenario and apply stride

6. Conclusion (0.5 Page)

- **Recap the Main Points:**
  - Security controls alone aren't enough.
  - Risk assessment models help, but traditional ones have gaps.
  - STRIDE and fuzzy logic can improve cloud security.
- **Future Directions:** AI-driven threat modeling, self-adaptive security frameworks.

Main goal of the paper is too prove:

- **Security controls alone** (e.g., encryption, firewalls) **don't fully protect cloud systems** from insider threats, new attack vectors, or log manipulation.

- **Traditional risk models** (e.g., CVSS, STRIDE, Attack Trees) what are they, pros, cons,
- **Fuzzy logic + STRIDE** helps **adapt to evolving threats** and improves risk prioritization

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